GEOG 172

Assessing Point Patterns

Let's use more information from the pattern and the space

Lecture 4-2

POINT PATTERNS

A set of *n* events at locations $S = [s_1, s_2, \ldots, s_n]$ in which each event (or point) s_i has location coordinates x_i, y_i, which occurs in a study region *a*.

PROCESS

A mathematical description (function) of the mechanism operating in the real world we believe to be responsible for a pattern.

QUESTION

Can you see a common shortcoming of both the *G* and *F* functions?

ANSWER: Both statistics use only the nearest neighbor for each event or location in a pattern. There is a lot more information available if you consider more neighbors in the pattern.

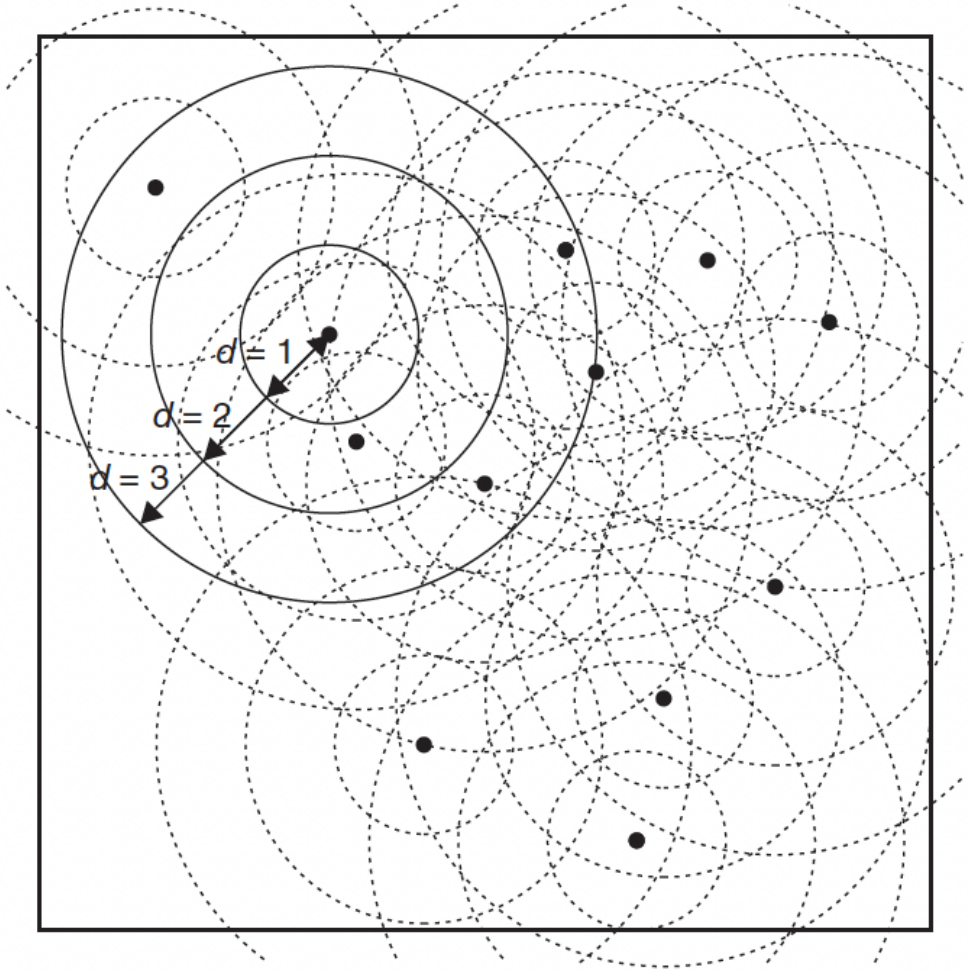
RIPLEY'S K FUNCTION

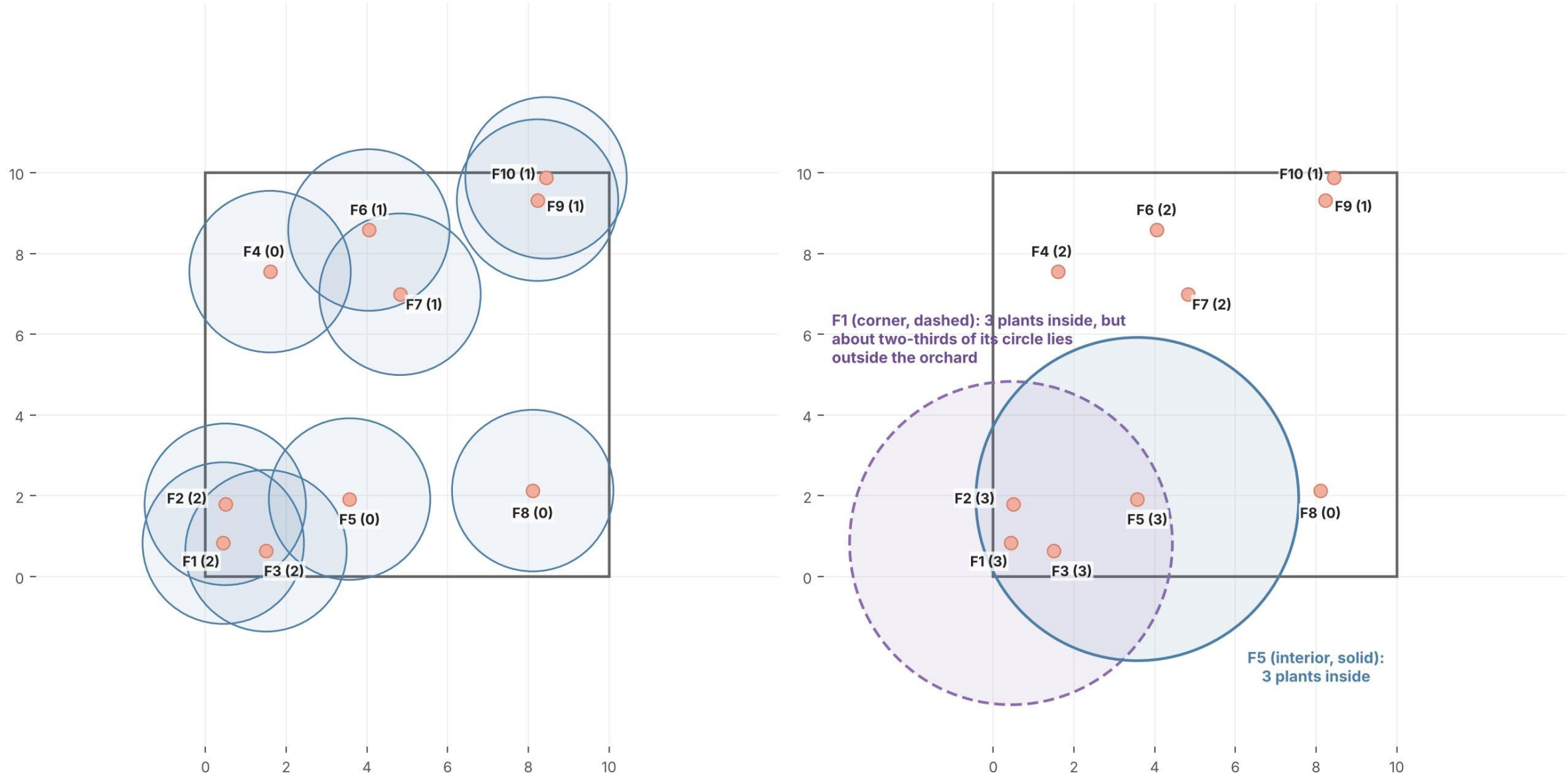
Considers distances to all neighbors

$$\widehat{K}(d) = \frac{R}{n^2} \sum_{i}^{n} \sum_{j=i}^{n} I_{ij}(d) .$$

where

- R is the area of the study region
- n is the number of points in S
- $I_{ij}(d)$ equal 1 if points i and j are closer than d





Each circle is centered on a lemon plant; the number in parentheses is how many OTHER plants lie inside it. The dark square is the official orchard; circle area outside it can never contain an observed plant.

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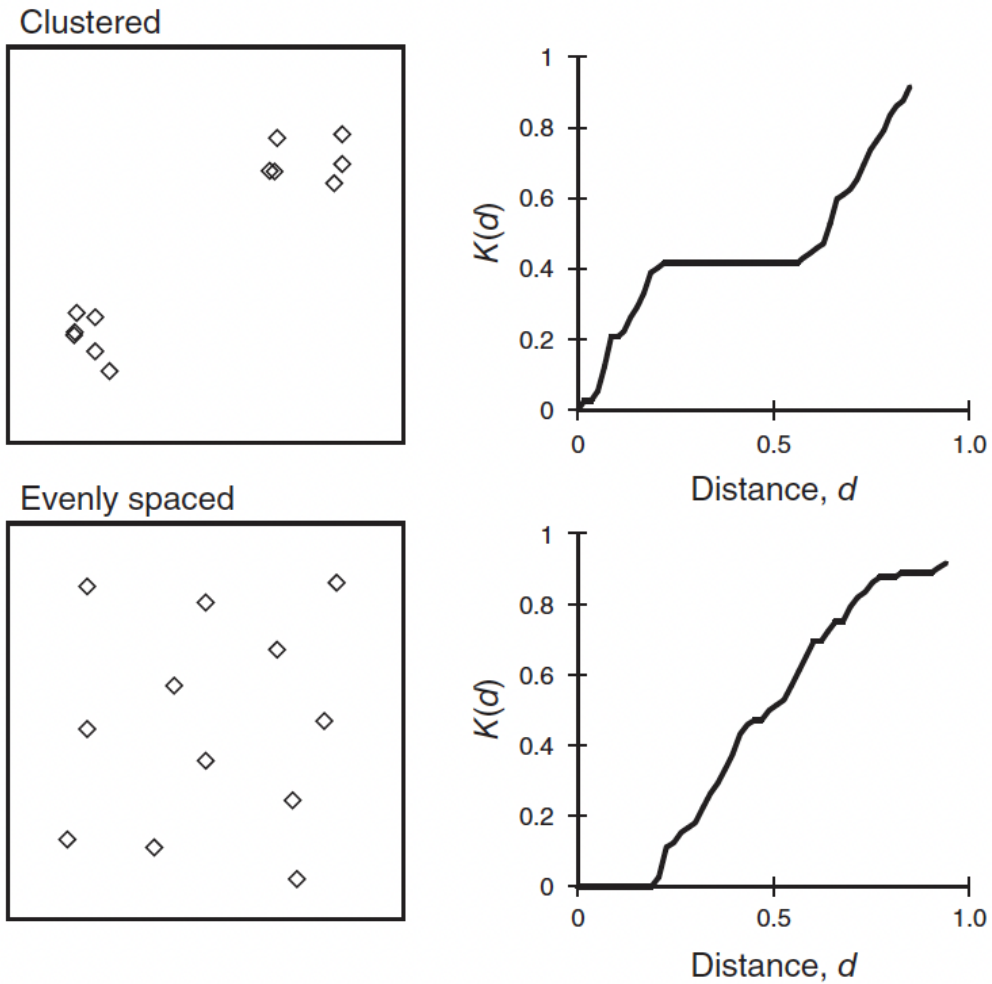


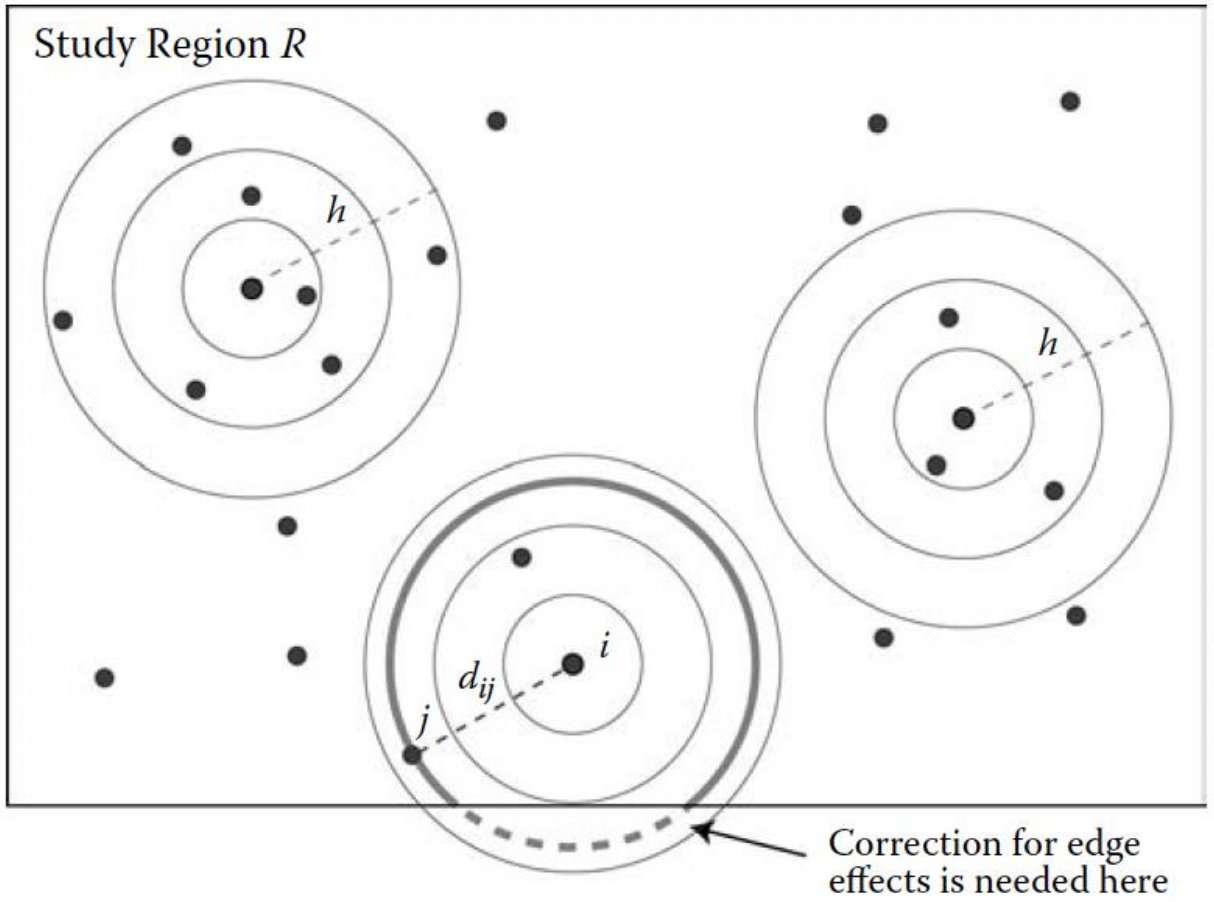
Figure 5.12 The *K* function for clustered and evenly spaced events.

EDGE-ADJUSTED RIPLEY'S K FUNCTION

Considers edges as well as distance

$$\widehat{K}_{edge}(d) = \frac{R}{n^2}\sum_{i}^{n}\sum_{j=i}^{n}\frac{I_{ij}(d)}{w_{ij}}$$

where w_{ij} is the proportion of the circumference of the circle centered on *i* that lies in the study region



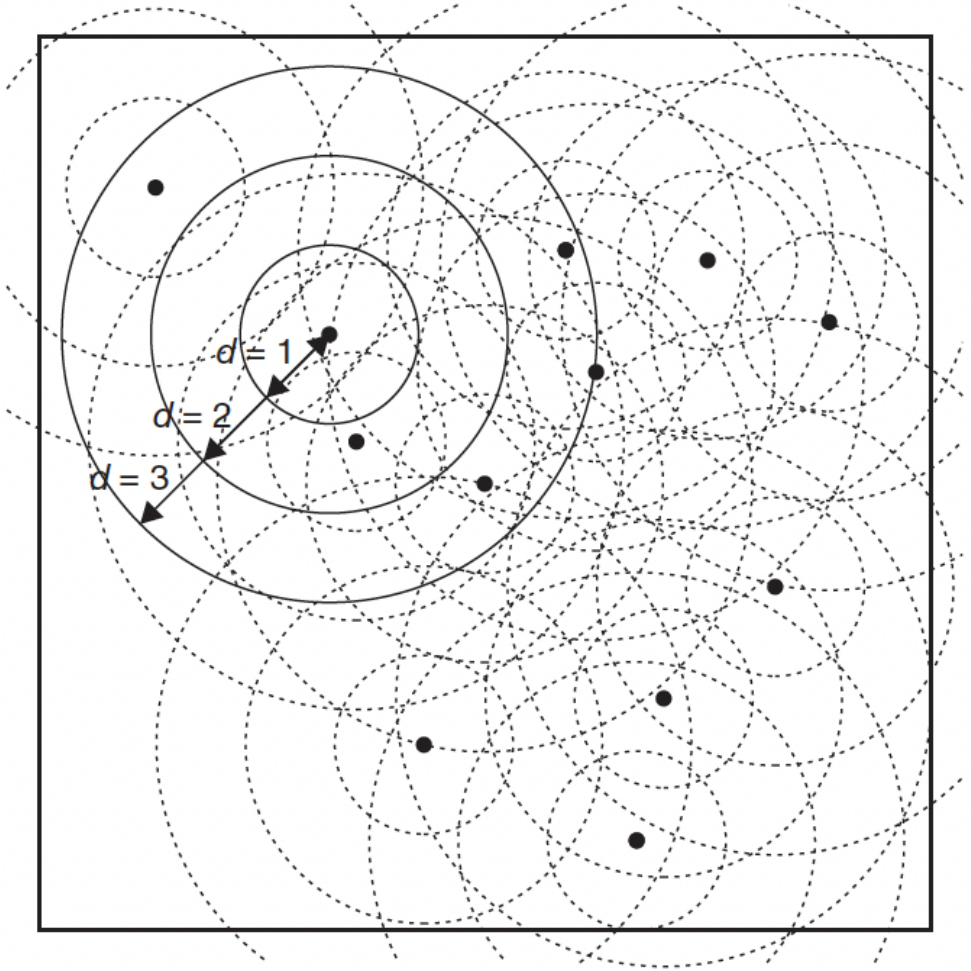
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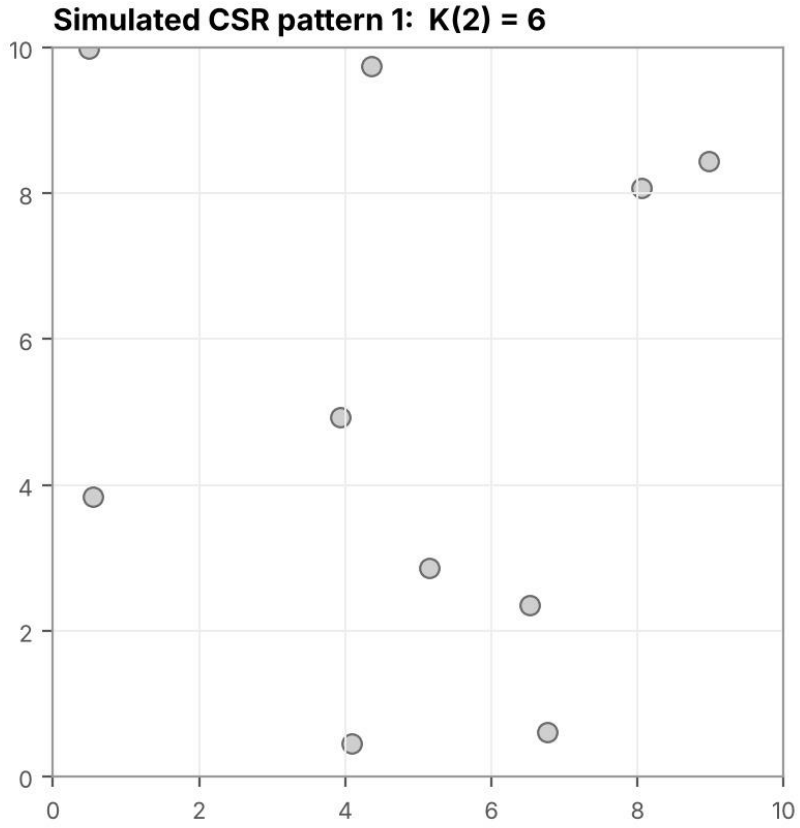
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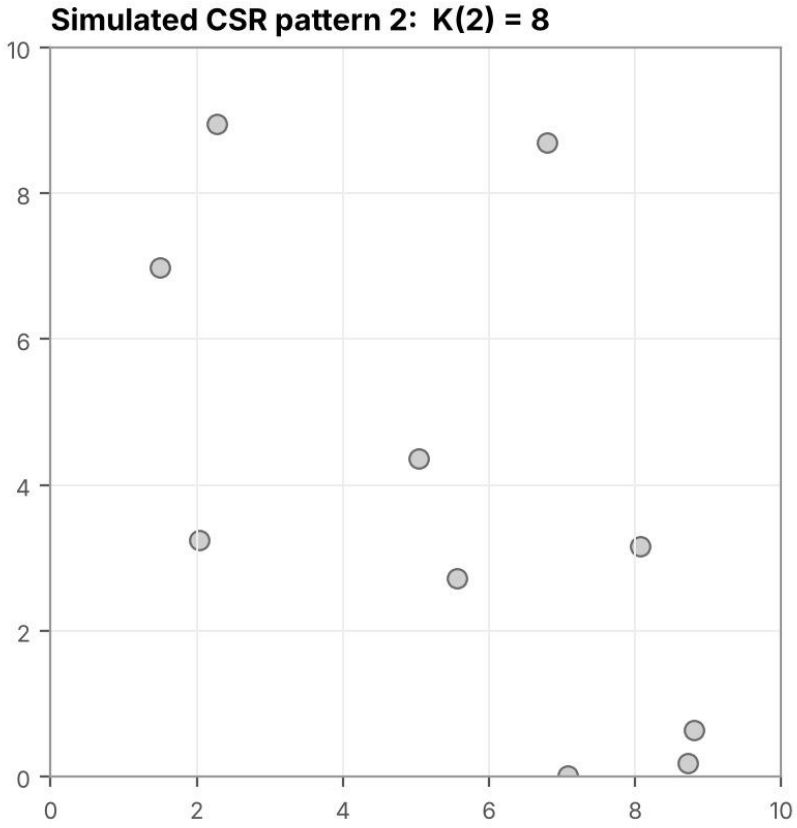
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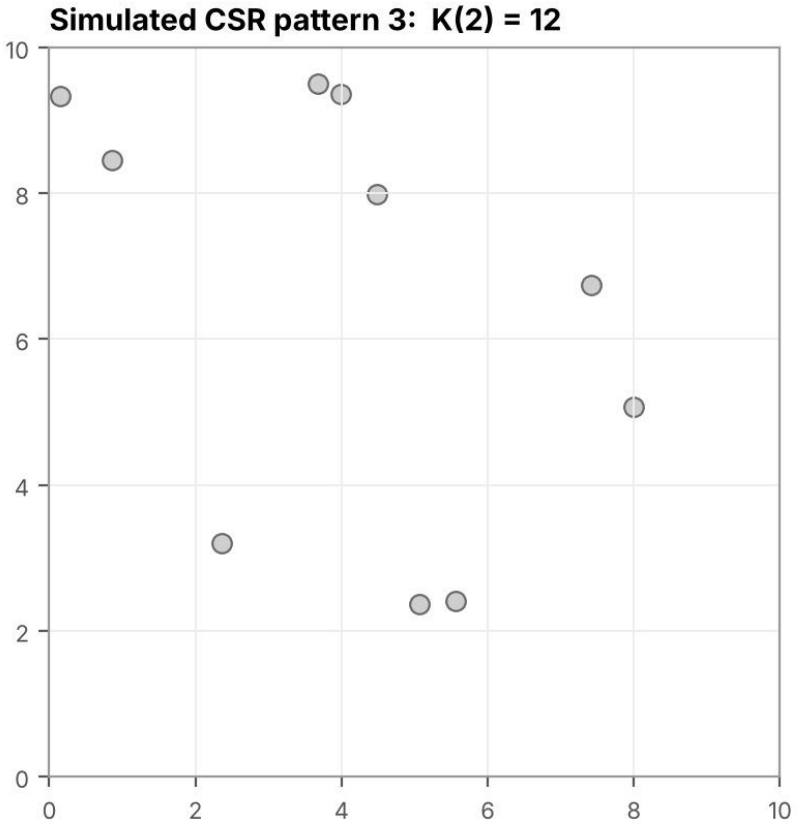
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Simulated CSR pattern 1: K(2) = 6
0
2
4
6
8
10


Simulated CSR pattern 2: K(2) = 8
0
2
4
6
8
10


Simulated CSR pattern 3: K(2) = 12
0
2
4
6
8
10

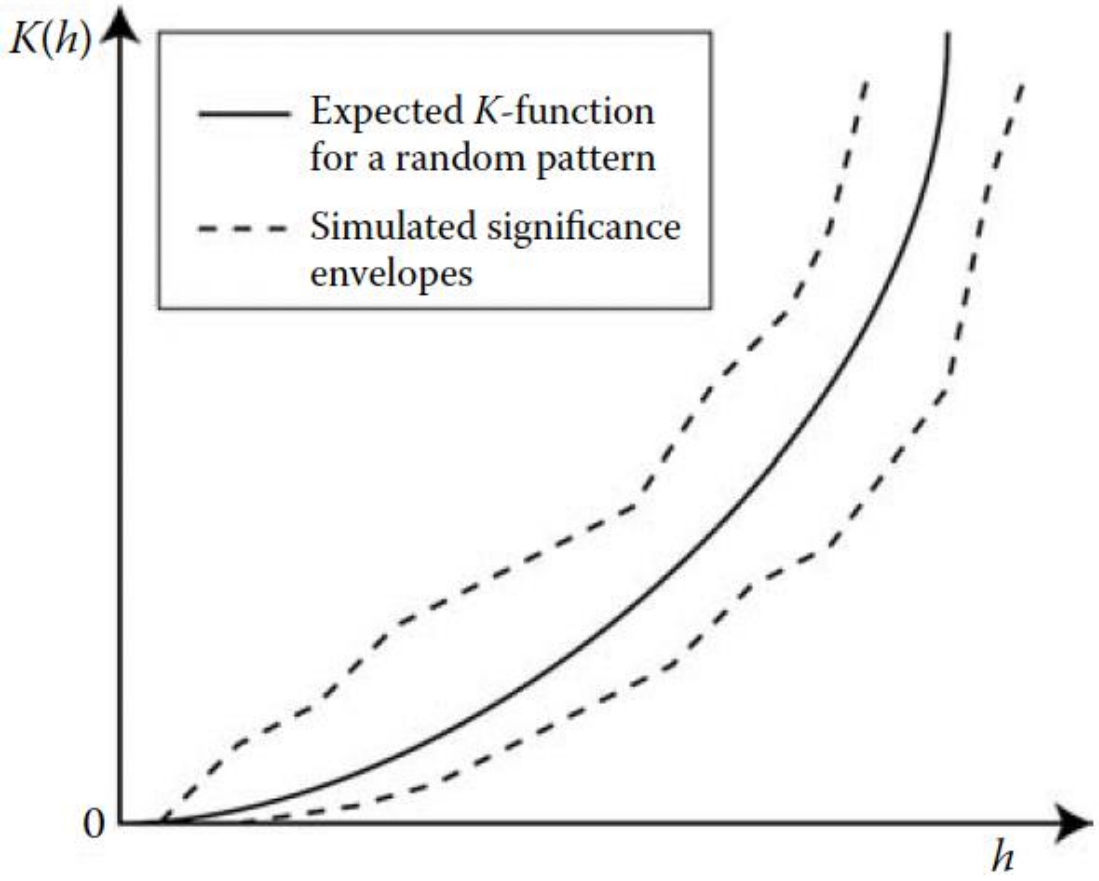
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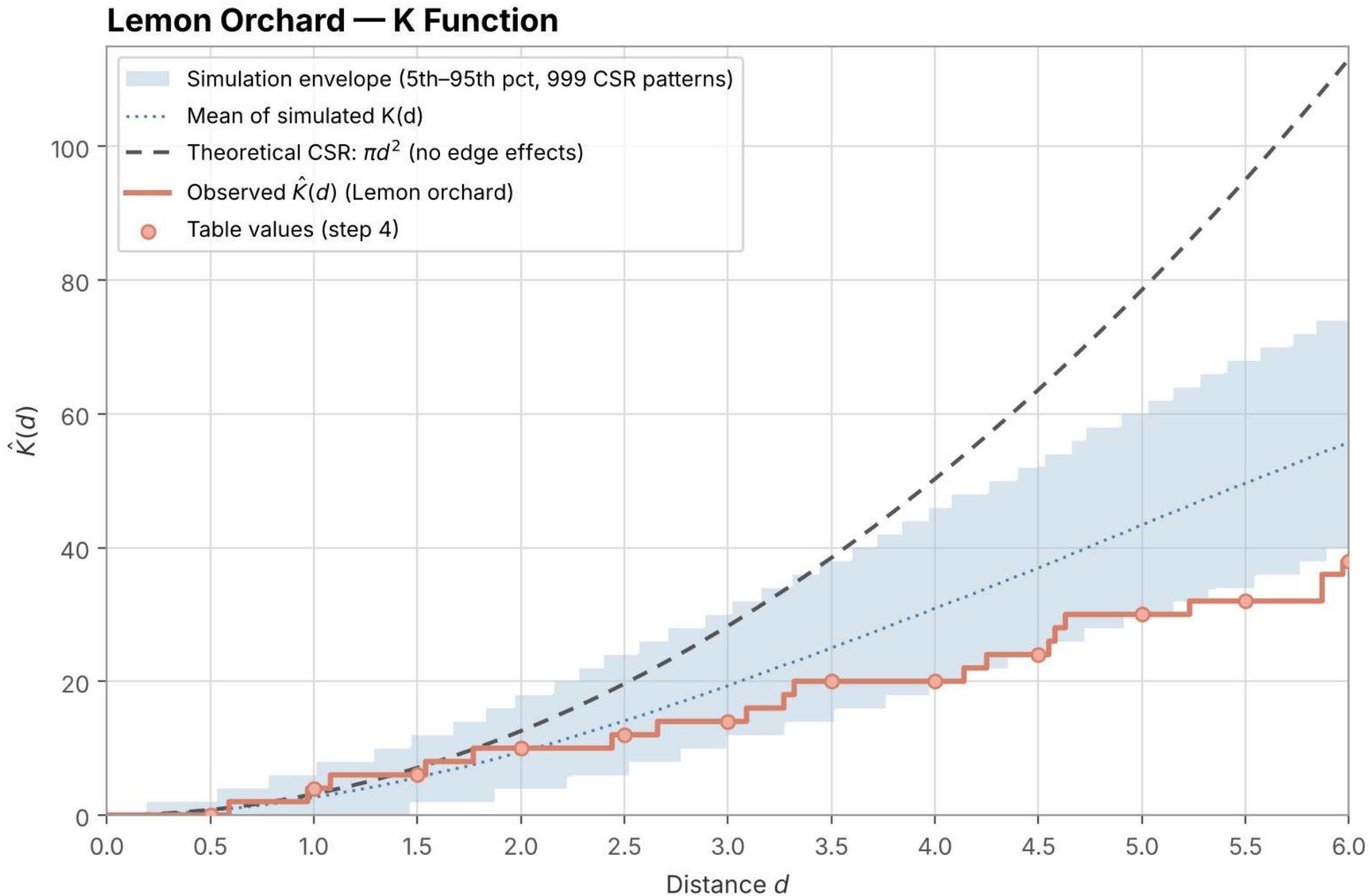
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Lemon Orchard — K Function
Simulation envelope (5th–95th pct, 999 CSR patterns)
Mean of simulated K(d)
Theoretical CSR: πd² (no edge effects)
Observed K̂(d) (Lemon orchard)
Table values (step 4)
K̂(d)
0
20
40
60
80
100
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
Distance d

Linking Pattern and Process

From concept to computer

SPATSTAT

Functions for describing a point pattern

Statistic	Example	Key arguments
Quadrat counts	quadratcount(X, nx = 8, ny = 8)	**nx**, **ny**: number of quadrats across and up the window's enclosing rectangle
NN distances	nndist(X)	**X**: the point pattern; returns each event's distance to its nearest neighbor
G function	Gest(X, r = d_values, correction = "rs")	**r**: distances to evaluate **correction**: "rs" (border), "km", "han", "none"
F function	Fest(X, r = d_values, correction = "rs")	Distances run from a fine grid of locations to the nearest event
K function	Kest(X, r = r_values, correction = c("none", "isotropic"))	**r**: distances to evaluate **correction**: "isotropic" (Ripley), "border", "translate", "none"

X is the point pattern, a ppp object. Every function here comes from the spatstat package.

R FUNCTIONS

```r
quadratcount(X, nx = 8, ny = 8)
```

- `X`: the point pattern (a `ppp` object)
- `nx`, `ny`: the number of quadrats across and up the window's enclosing rectangle

```r
nndist(X)
```

- `X`: the point pattern; returns each event's distance to its nearest neighbor

R FUNCTIONS

```r
Gest(X, r = d_values, correction = "rs")
```

- `r`: the distances at which G is evaluated
- `correction`: `"rs"` (border), `"km"` (Kaplan–Meier), `"han"` or `"none"`

```r
Fest(X, r = d_values, correction = "rs")
```

- `r`: the distances at which F is evaluated
- `correction`: `"rs"` (border), `"km"`, `"cs"` or `"none"`
- `eps`: the spacing of the grid of locations F is measured from (spatstat's default here)

R FUNCTIONS

```
r

Kest(X, r = r_values, correction = c("none", "isotropic"))
```

- `r`: the distances at which K is evaluated
- `correction`: `"none"`, `"isotropic"` (Ripley), `"translate"` or `"border"`
- `rmax`: the largest distance, if `r` isn't given

R FUNCTIONS

```r
envelope(X, Kest, nsim = 99, nrank = 5, correction = "border",
         simulate = expression(rpoispp(lambda_im)[win]))
```

- second argument (`fun`): the summary function to compute, such as `Kest`, `Gest` or `Fest`
- `nsim`: the number of simulated patterns
- `nrank`: which ranked simulated value forms each edge of the envelope (5 of 99 gives roughly the 5th and 95th percentiles)
- `correction`: passed on to `fun`
- `simulate`: how each null pattern is generated; if omitted, CSR in the same window

TUTORIAL

Robbery is one of the crimes residents cite most often when they describe a neighborhood as unsafe. Community groups argue that robberies are concentrated in a handful of neighborhoods and call for more policing and targeted patrols. The police department may reply that robberies simply happen more frequently where people live. Both sides are looking at the same map, but they disagree about the process that produced it.

Your task this week is to weigh these two positions using the robberies reported in Chicago during 2025. This tutorial asks: **Do reported robberies in Chicago during 2025 form a clustered point pattern, and if they do, is the clustering explained by where people live?**

TAKEAWAYS

- Point Pattern Statistics (NN, G, F, K) all use different parts of the information contained in the data to measure whether randomness is or is not present.
- Ripley's K examines the scale at which different spatial patterns may be dominant
- These statistics can be sensitive to edge effects

DIFFERENCE IN RIPLEY'S K FUNCTION

Compare the spatial pattern of cases to the spatial pattern of controls

$$\widehat{K}(d) = \frac{R}{n^2} \sum_{i}^{n} \sum_{j=i}^{n} I_{ij}(d) .$$

$$K_{case}(d) - K_{control}(d)$$

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$$K_{case}(d) - K_{control}(d)$$

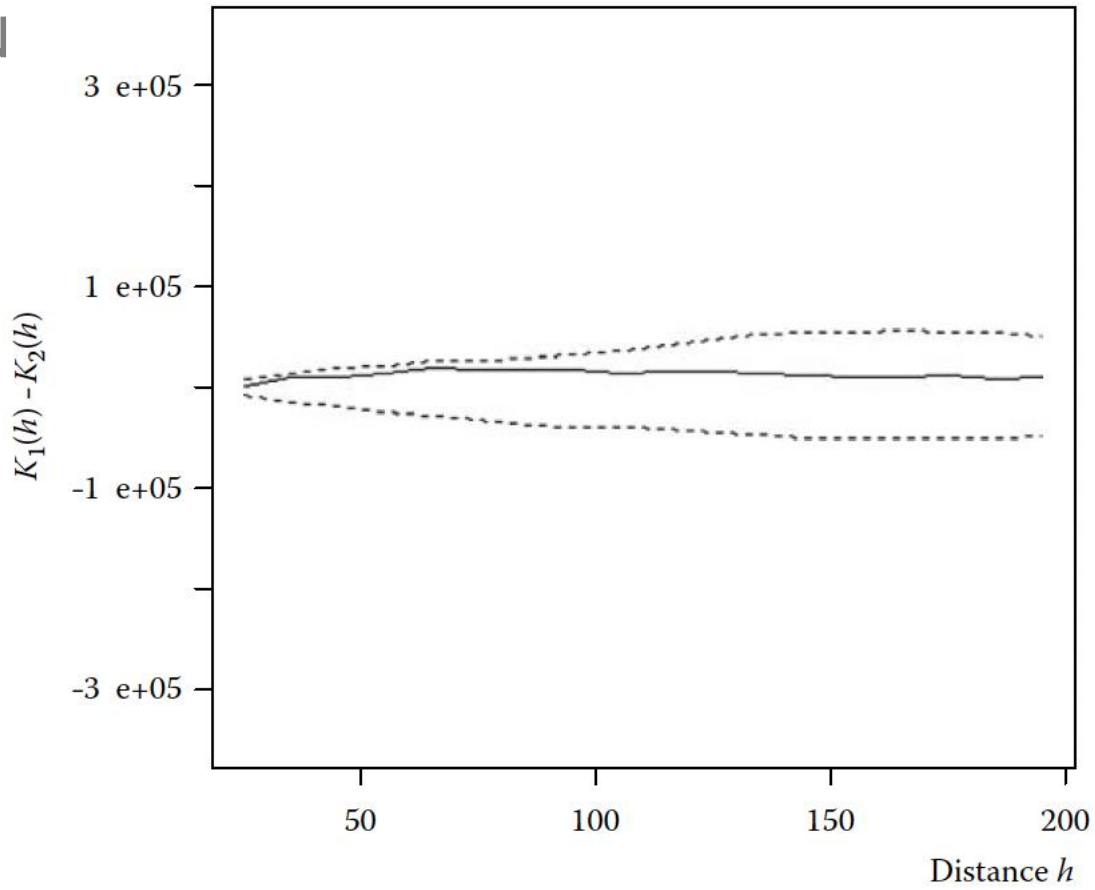


FIGURE 2.12
Difference between *K*-functions for cases and controls for Humberside leukemia data.